

Evaluation Choices Determine Apparent Quantum Kernel Robustness under Distribution Shift

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Abstract

Quantum-kernel robustness claims depend on how candidates are tuned and selected. We test this dependence in a controlled kernel-swap benchmark spanning four security scenarios, three shift mechanisms, two classifiers and fifteen pipeline realizations per setting. Under same-test OOD-oracle selection and fixed regularization, fidelity kernels appear to improve balanced accuracy over linear and radial-basis-function baselines by up to **+0.037**. When regularization is tuned from training data, selection uses no OOD labels and candidate budgets are matched, the dataset-equal quantum-minus-classical effect becomes **−0.005** for support-vector classification and **−0.002** for Gaussian-process classification. The primary support-vector result is classical-favoured or conditionally tied in seven of eight scenario-groups. At matched effective rank, classical kernels are at least as accurate in every group. Effective rank and OOD kernel-target alignment are associated with accuracy but favour neither family, while finite-shot estimation distorts exact-statevector effective rank. Evaluation therefore removes the apparent robustness advantage within the low-qubit regimes studied.

Keywords: quantum machine learning; quantum kernels; distribution shift;
out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian
processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods embed classical data into quantum states and use state overlaps as similarities for otherwise classical learners [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). This separation makes them a clean setting for asking whether quantum-induced feature geometry provides useful inductive bias. It also makes comparison unusually sensitive to evaluation design: the downstream learner can be held fixed, but conclusions still depend on the encoding, bandwidth, regularization, classical reference set, and rule used to select a candidate [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#).

Recent benchmarks have accordingly shifted the question from whether an isolated quantum model can perform well to whether a claimed gain survives strong and symmetric controls. Broad QML studies find competitive classical baselines once pipelines are aligned [Bowles et al. \(2024\)](#); quantum-kernel benchmarks show substantial sensitivity to feature map and hyperparameter choice [Schnabel and Roth \(2025\)](#); and a concurrent tabular study using nested cross-validation, spectral analysis, and hardware validation finds no significant quantum win across its tested pairs [Kakavand et al. \(2026\)](#). These studies evaluate predominantly identically distributed train–test settings. They do not address a further deployment question: what happens when the target distribution changes and its labels cannot be used to choose the deployed kernel?

Distribution shift makes model selection part of the estimand. Work on domain generalization and real-world shift has shown that robustness gains can disappear when search budgets, hyperparameters, and selection criteria are standardized [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#); [Zhou et al. \(2023\)](#). A selector that inspects the shifted test labels estimates an oracle upper bound, not deployable performance; a family with more candidates has more opportunities to win that selection. Fair quantum–classical evaluation under shift must therefore control

not only the classifier and data split, but also regularization, candidate budget, and label access.

Security detection provides a useful stress test because shift is intrinsic to the application. Malware evolves over time, and experimental bias can overstate detector performance [Anderson and Roth \(2018\)](#); [Pendlebury et al. \(2019\)](#); [Arp et al. \(2022\)](#); network traffic changes with infrastructure and attack campaigns [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Quantum classifiers have already been evaluated on intrusion-detection benchmarks, sometimes with reported gains over small classical reference sets on i.i.d. splits [Abreu et al. \(2024, 2025\)](#); [Bellante et al. \(2025\)](#). We use these datasets not to propose a security system, but to expose kernel families to reproducible constructed shifts and to a held-out attack-campaign shift.

The closest robustness-specific concurrent work studies additive input corruption and proposes a spectral phase encoding [Herrero Gómez et al. \(2026\)](#). Our question is complementary: rather than optimize a new encoding, we audit how evaluation choices change a family-level conclusion under constructed and held-out-campaign shifts. The distinguishing axes are no-OOD-label selection, equal candidate budgets, per-configuration train-only regularization, fixed-case inference, and a geometry analysis tied specifically to behaviour after the distribution moves.

We implement a controlled *kernel-swap* benchmark in which preprocessing, splits, classifier, and family-internal selection logic are fixed while only the kernel changes. Four security scenarios from EMBER, UNSW-NB15, and ToN-IoT yield eight scenario-groups across constructed and held-out-campaign shifts. Support-vector and Laplace Gaussian-process classifiers consume the same precomputed Gram matrices. The quantum family contains four fidelity feature-map families with angle-scale sweeps; the classical family extends linear and RBF references with polynomial, Laplacian, and Matérn kernels and symmetric bandwidth freedom. Each configuration’s regularization is tuned from training data alone. The primary deployed candidate is

selected on a disjoint ID-validation split, and the two extended families receive equal candidate budgets. Fifteen pipeline realizations per setting quantify conditional variability; the benchmarks are treated as fixed case studies rather than as random draws from an undefined population.

The comparison is coupled to a mechanistic audit. We measure effective rank, centered kernel-target alignment, concentration, and geometric difference [Roy and Vetterli \(2007\)](#); [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#); [Huang et al. \(2021\)](#); use partial association, leave-one-dataset-out prediction, and effective-rank matching to restrict interpretation; and perturb fidelity estimates with finite shots to test whether exact statevector geometry survives measurement.

The evaluation choices change the conclusion. With same-test oracle selection, fixed regularization, and customary linear+RBF baselines, fidelity kernels appear to improve OOD balanced accuracy by as much as +0.037. With train-only regularization, no-OOD-label selection, and equal candidate budgets, the dataset-equal quantum-minus-classical delta is -0.005 for SVC and -0.002 for the GP classifier. The primary SVC comparison is classical-favoured or conditionally tied in seven of eight scenario-groups. Effective rank and OOD alignment remain associated with accuracy, but the association is regime-dependent and shared by both families; at matched effective rank, the classical configuration is at least as accurate in every group. Finite-shot estimation inflates effective rank and destabilizes candidate selection.

The contribution is therefore an evaluation result rather than an advantage claim: it identifies which protocol choices create the optimistic quantum-kernel robustness result, supplies a deployable equal-budget comparison under shift, and shows that the observed geometry does not uniquely identify quantum provenance. The scope is deliberately limited to the low-qubit fidelity kernels, datasets, and shift constructions studied here.

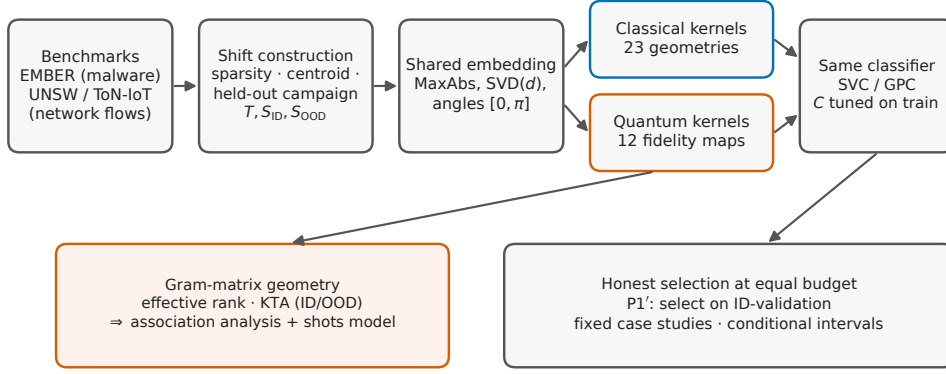


Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding, receive per-configuration regularization tuned on the training split, and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by honest, equal-budget selection on an in-distribution validation split (P1'), summarized as fixed case studies with conditional intervals; the Gram matrices feed the geometry-association analysis and the finite-shot model.

2 Results

2.1 Overview

We report results under the controlled kernel-swap protocol described in Section 4. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a scenario-group; the only varying component is the kernel, and every number below aggregates fifteen pipeline realizations per setting. The argument proceeds in three movements. First, under test-peeking oracle selection and fixed regularization, the fidelity kernels show an apparent OOD advantage over linear and RBF baselines (Section 2.2). Second, we tune each configuration on training data alone, select on an in-distribution validation split disjoint from both test splits, and match the candidate budgets of the extended families. Under this controlled comparison no robust family-level advantage remains; the primary SVC comparison is classical-favoured or tied in seven of eight scenario-groups (Section 2.4). Third, a geometry analysis shows that effective rank and OOD alignment are associated with—but do

not causally determine—performance, that the association is regime-dependent, and that it favours neither family (Section 2.5); a finite-shot analysis then shows the exact geometry does not transfer unchanged to finite measurement (Section 2.6). Throughout, we treat the eight scenario-groups as fixed case studies and report effect sizes with conditional intervals rather than a population p -value, because the settings within a benchmark share data pools and are not exchangeable (Section 4).

2.2 The apparent advantage under oracle selection

We begin with the comparison that dominates the literature: fidelity-based quantum kernels against linear and RBF baselines, with each family’s representative chosen by Best-by-OOD selection—the configuration with the highest balanced accuracy on the OOD test split itself—and the classical kernels left at the default regularization $C = 1$. Under these conventions, on EMBER, the quantum family shows an OOD advantage of +0.037 under SVC and +0.028 under the GP classifier against linear+RBF, shrinking to +0.004/+0.002 once the classical family is enlarged with heavier-tailed kernels. Supplementary Table 2 gives the complete oracle results.

Two features favour this apparent quantum result. Best-by-OOD is an *oracle*: it uses the labels it then reports. Fixing $C = 1$ also leaves classical candidates at a potentially unsuitable regularization point. These values are therefore descriptive upper bounds, not deployable effects.

2.3 A specification curve locates the reversal

To separate the contribution of each evaluation choice from the choice of a preferred endpoint, we enumerated all ten logically valid, already-computed specifications fixed before this extension (Figure 2). The specifications vary four axes in a prespecified order: same-test OOD-oracle versus ID-based selection, fixed versus train-cross-validated SVC regularization, customary linear/RBF versus extended classical

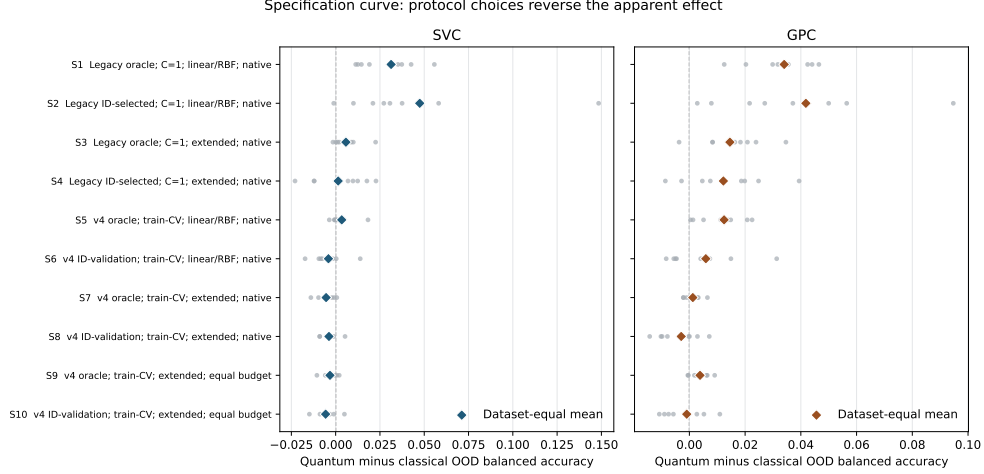


Fig. 2 A frozen specification curve shows that evaluation choices reverse the apparent family effect. Grey points are the eight security scenario-groups; diamonds are dataset-equal means. Specifications are shown in their prespecified S1–S10 order, not sorted by effect. They vary target-label access, SVC regularization, classical reference strength, and candidate budget. Positive values favour the fidelity-kernel family. GPC has no C -regularization axis. No specification-level hypothesis test is performed.

references, and native versus equal candidate budgets. Every specification retains all eight security scenario-groups.

The family-level conclusion spans both sides of zero. For SVC, the dataset-equal quantum-minus-classical mean ranges from $+0.047$ under legacy ID selection with fixed $C = 1$ and the customary reference (S2) to -0.006 under ID-validation selection, train-only regularization, the extended reference, and equal budgets (S10). For the GP classifier, for which the regularization axis is not applicable, the corresponding range is $+0.042$ to -0.003 . Expanding the classical reference contracts the positive effect before honest selection or budget matching is imposed; train-only SVC regularization and ID-validation selection then move the controlled estimates to zero or slightly negative. The path is not monotone in every intermediate specification, which is precisely why reporting only an oracle and a final controlled endpoint would be insufficient. The curve shows that the apparent advantage is conditional on a bundle of evaluation decisions rather than a stable property of quantum provenance.

323 2.4 A fair comparison removes the advantage

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325 A deployable comparison must tune each configuration’s regularization without look-
326 ing at the test data, select the deployed configuration without OOD labels, and
327 compare families at the same candidate budget. We implement all three (Section 4).
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329 For every kernel candidate, the SVC regularization C is chosen by five-fold cross-
330 validation on the training Gram matrix alone. The deployed configuration is then the
331 one with the best balanced accuracy on an in-distribution *validation* split, held out
332 from the ID test split (protocol P1’); no OOD label is ever consulted in selection.
333
334 The primary comparison uses the same candidate budget for both extended fami-
335 lies (60 per family in full-coverage groups and 20 in reduced-coverage groups), with
336 the full classical pool retained as a sensitivity. We treat the eight scenario-groups
337 as fixed case studies and report, per group, the quantum-minus-classical OOD delta
338 with a conditional interval over the five split realizations, and no population p -value
339 (Section 4).
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341

342 Table 1 and Figure 3 give the result: no robust advantage remains against either
343 reference. Against the equal-budget extended classical family, the P1’ delta is classical-
344 favoured or within the conditional uncertainty of zero in seven of the eight scenario-
345 groups under the primary SVC comparison, with a dataset-equal-weighted mean of
346 -0.005 (leave-one-dataset-out range $[-0.006, -0.004]$); under the GP classifier the
347 mean is -0.002 ($[-0.003, +0.000]$). Against the customary linear+RBF baseline—the
348 comparison that produced the $+0.037$ oracle result of Section 2.2—the no-OOD-label
349 delta is -0.004 under SVC and $+0.004$ under the GP classifier, with heterogeneous
350 signs across scenario-groups. The single regime that retains a small quantum lean
351 under both equal-budget comparisons is UNSW-NB15 reconnaissance under held-out
352 attack-campaign shift ($+0.005$ for both classifiers); both conditional intervals include
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Table 1 No-OOD-label selection results (eight scenario-groups, fifteen pipeline realizations per setting). Quantum-minus-classical OOD balanced-accuracy delta under P1' (select on ID-validation, tune per-configuration C on train, evaluate on OOD test). The extended-family comparison is the frozen equal-budget analysis: 60 candidates per family in full-coverage groups and 20 in reduced-coverage groups. Its interval is conditional pipeline-realization uncertainty over the five split seeds. The customary linear+RBF comparison is contextual rather than an equal-budget endpoint; the full-pool extended-family sensitivity remains in the machine-readable artifact. No population p -value is reported.

Scenario	Clf.	vs lin./RBF	vs ext. (equal B)	conditional interval
EMBER $m1$	SVC	-0.0095	-0.0058	$[-0.0340, +0.0223]$
	GPC	-0.0050	-0.0009	$[-0.0101, +0.0083]$
EMBER $m2$	SVC	+0.0139	-0.0064	$[-0.0137, +0.0009]$
	GPC	+0.0149	+0.0027	$[-0.0027, +0.0081]$
UNSW-DoS (campaign)	SVC	-0.0043	-0.0033	$[-0.0139, +0.0073]$
	GPC	+0.0041	-0.0057	$[-0.0120, +0.0005]$
UNSW-DoS (constructed)	SVC	-0.0085	-0.0044	$[-0.0119, +0.0031]$
	GPC	-0.0056	-0.0108	$[-0.0177, -0.0039]$
UNSW-Recon (campaign)	SVC	+0.0004	+0.0050	$[-0.0058, +0.0157]$
	GPC	+0.0074	+0.0052	$[-0.0019, +0.0123]$
UNSW-Recon (constructed)	SVC	-0.0076	-0.0089	$[-0.0146, -0.0031]$
	GPC	-0.0083	-0.0075	$[-0.0208, +0.0059]$
ToN-IoT (campaign)	SVC	-0.0172	-0.0148	$[-0.0268, -0.0029]$
	GPC	-0.0046	-0.0090	$[-0.0190, +0.0010]$
ToN-IoT (constructed)	SVC	-0.0016	-0.0011	$[-0.0049, +0.0028]$
	GPC	+0.0313	+0.0109	$[+0.0017, +0.0202]$
Dataset-equal mean, SVC		-0.0043	-0.0050	$[-0.0060, -0.0040]$
Dataset-equal mean, GPC		+0.0043	-0.0019	$[-0.0028, +0.0003]$

zero. The full-pool extended-family sensitivity is similar in direction and scale (dataset-equal means -0.004 for both classifiers), so the conclusion does not depend on giving the classical family its larger pool.

Rank matching does not reveal a residual quantum advantage.

We pair each quantum configuration with the classical configuration of nearest training effective rank, within the same run and dimension, and compare their OOD accuracy. Figure 4 reports the outcome: in all eight scenario-groups the median rank-matched delta is negative (quantum minus classical, between -0.001 and -0.009), and the quantum configuration is the more accurate of the pair in only 22–37% of matches. Within

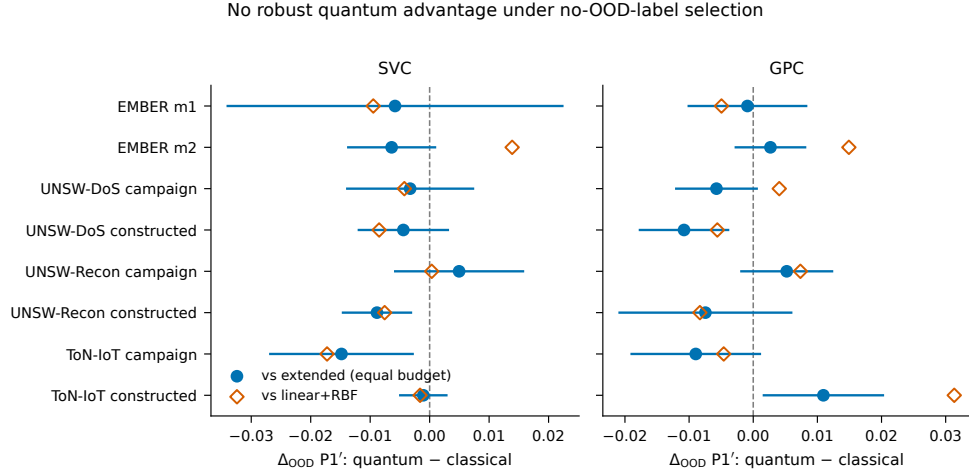


Fig. 3 No robust quantum advantage under no-OOD-label selection. Per-scenario-group P1' OOD deltas (quantum minus classical), for SVC (left) and the GP classifier (right). Filled circles compare against the equal-budget extended classical family with conditional pipeline-realization intervals; open diamonds compare against the customary linear+RBF reference. All eight scenario-groups are shown.

the overlap covered by this observational matching, classical kernels are therefore at least as accurate at comparable effective rank.

2.5 Geometry is associated with, but does not determine, robustness

The two families are close because both reach similar kernel geometry, and geometry—not quantumness—is what carries information about behaviour under shift. That information, however, is partial: it is a regime-dependent association, not the single governing law that optimistic accounts (our own earlier report included) proposed.

The geometry–OOD link is a regime-dependent association, not a law.

Two descriptors of the training and OOD Gram matrices carry information about which kernels generalize: the effective rank of the training kernel and the centered kernel-target alignment on the OOD split. But their association with OOD accuracy is regime-dependent, not a single mechanism. Within scenario-groups (Figure 5 and

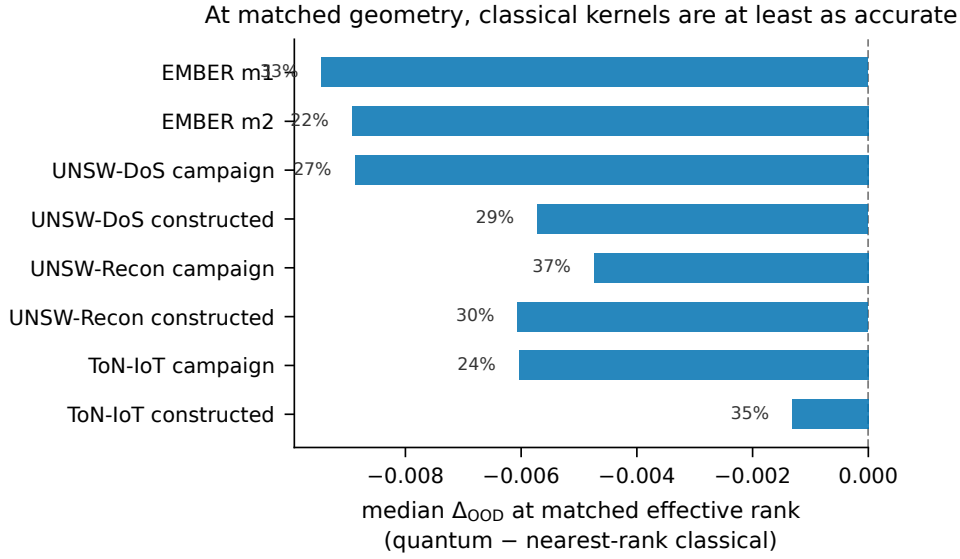


Fig. 4 At matched effective rank, classical kernels are at least as accurate. For each quantum configuration we find the classical configuration of nearest training effective rank (within run and dimension) and take the OOD balanced-accuracy difference; bars are per-scenario-group medians, annotated with the fraction of matched pairs in which the quantum configuration is the more accurate.

Supplementary Table 3), the median Spearman correlation between effective rank and OOD accuracy ranges from +0.80 (EMBER *m2*) down to −0.15 (UNSW-NB15 reconnaissance under the constructed shift); partial correlations that control for family label and length scale track the raw values closely. OOD alignment is more consistently positive (+0.22 to +0.64) but likewise varies by regime. A geometry model fit on three datasets and used to rank the held-out dataset’s kernels achieves only Spearman 0.23–0.40. We therefore describe a genuine but partial association, not a causal mechanism or deployable predictor.

2.6 Finite-shot estimation: the geometry is a statevector idealization

All quantum kernels above are computed by exact statevector simulation, which is the right idealization for a geometry study but not what a device would return. To

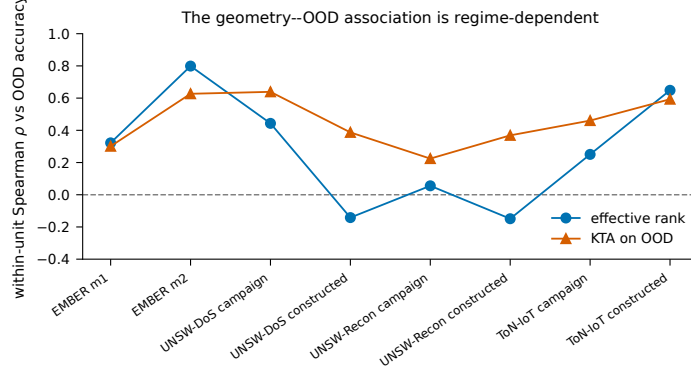


Fig. 5 The geometry-OOO association is regime-dependent. Median within-unit Spearman correlation between each geometry descriptor and OOD balanced accuracy, per scenario-group (SVC). Effective rank orders accuracy strongly in some regimes (EMBER *m2*, ToN-IoT *m2*) and weakly or negatively in others (the UNSW constructed-shift scenarios); OOD alignment is more consistently positive but never uniform. No single descriptor governs all regimes.

probe how the picture degrades away from that idealization we apply a *finite-shot fidelity-estimation perturbation model*: each fidelity entry is replaced by a binomial estimate from s shots, train blocks are symmetrized, unit-diagonal, clipped to $[0, 1]$, and projected to the positive-semidefinite cone (Section 4). This is not a hardware simulation—no device noise, mitigation, or transpilation is modelled—but it isolates the statistical cost of estimating a fidelity from finite measurements. Table 2 reports the effect on the $q1000$ subset. OOD alignment and OOD accuracy are robust: median KTA_{OOD} deviates by at most 0.0004 from the exact value even at 128 shots, and OOD balanced accuracy by 0.018–0.033. Effective rank is not: finite sampling inflates it by a median factor of 2.1 at 128 shots, falling to 1.15 only at 8192 shots, because sampling noise adds spurious spectral mass. The within-family configuration selected by ID-validation changes identity under perturbation (agreement 7–21%), but the accuracy cost of that instability is negligible (mean regret ≤ 0.003), so the flatness of the family optimum—already visible in the exact results—persists. The practical reading: the alignment description we can measure transfers to finite shots, but the effective-rank description, and any story that leans on it, is a statevector-exact quantity that would need many shots to reproduce.

Table 2 Finite-shot fidelity-estimation perturbation (EMBER+network $q1000$ subset, SVC). “OOD acc. dev.” is the median absolute deviation of OOD balanced accuracy from the statevector-exact value; “KTA dev.” the same for OOD alignment; “rank ratio” the median ratio of perturbed to exact effective rank; “sel. agree” the fraction of runs whose ID-validation-selected configuration is unchanged; “regret” the mean OOD accuracy lost by deploying the perturbed selection.

Shots	OOD acc. dev.	KTA dev.	rank ratio	sel. agree	regret
128	0.0327	0.0004	2.12	7%	+0.0007
512	0.0273	0.0001	1.56	10%	+0.0017
2048	0.0240	0.0001	1.29	19%	−0.0025
8192	0.0181	0.0000	1.15	21%	−0.0001

2.7 Predictive uncertainty under shift

The Laplace GP probabilities provide a secondary uncertainty check (Supplementary Figure 1). For the honestly selected candidate, predictive entropy rises from ID to OOD for both families in every scenario-group, while expected calibration error changes moderately and comparably. Because neither family has an accuracy advantage, the fidelity kernels show no calibration benefit unavailable to classical kernels. We do not interpret rising entropy alone as evidence of calibration.

3 Discussion

The study’s central interpretive move is to replace the question “do quantum kernels beat classical kernels under shift?” with “which properties of a kernel are associated with its behaviour under shift, and does either family have exclusive access to them?”. Read through that lens, the results are clear and negative. The apparent advantage over linear and RBF baselines does not survive the combined correction of oracle selection, fixed regularization, and unequal candidate families. It does not reappear against the equal-budget extended family or at matched effective rank. The geometry descriptors carry information as regime-dependent associations rather than a governing law, and they are shared: a well-tuned heavy-tailed classical kernel occupies the same

599 observed geometric regime as the fidelity maps. On present evidence there is no robust
600 out-of-distribution advantage for fidelity kernels in the low-qubit settings studied,
601 and the only residual—a modest in-distribution separability edge under the proba-
602 bilistic classifier—is small, classifier-dependent, and itself shrinks under no-OOD-label
603 selection.
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605
606 From the perspective of the distribution-shift literature, this reinforces a general
607 lesson: conclusions about robustness are only as credible as the protocol used to obtain
608 them [Quiñonero-Candela et al. \(2008\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). Our results add a QML-specific corollary. The strength of
609 the classical baseline, whether its regularization is tuned, whether model selection
610 may consult the target labels, and the size of each family’s candidate budget are
611 not implementation details: each can move the quantum-versus-classical verdict, and
612 together they turn an apparent +0.037 advantage into a null. A demonstration that
613 fixes any of them in the quantum family’s favour, reported alone, would mislead—and
614 much of the optimistic quantum-kernel literature fixes several at once.
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616 In-distribution separability is the one axis on which the fidelity kernels retain any
617 edge, and even it must be stated carefully. Under the oracle that selects and evaluates
618 on the same in-distribution split, the quantum family separates the classes slightly bet-
619 ter than linear+RBF in every EMBER setting; but under honest selection—choosing
620 on a validation split and reporting on a disjoint test split—the edge falls to +0.003
621 under SVC (mixed in sign across groups) and +0.012 under the probabilistic classifier,
622 and it does not translate into any out-of-distribution benefit. This ID–OOD dissocia-
623 tion is real but modest, and it cuts against, not for, an operational recommendation:
624 a family that is marginally more separable in-distribution yet no more robust under
625 shift offers nothing to a deployment whose whole difficulty is the shift. The pattern is
626 consistent with the “accuracy on the line” literature [Miller et al. \(2021\)](#), in which ID
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and OOD accuracy usually move together; here they decouple weakly, and geometry—
shared by both families—rather than quantumness marks the axis along which they
do.

The geometry association connects several strands of literature. That length scale
is decisive for kernel behaviour is known for both families [Shaydulin and Wild \(2022\)](#);
[Canatar et al. \(2023\)](#); our contribution is the comparative consequence—once length
scale is tuned symmetrically and regularization is tuned at all, the classical fam-
ily reaches the same geometric regime, so asymmetric tuning does not merely bias
quantum-versus-classical comparisons, it manufactures the apparent advantage. The
geometric difference of Huang et al. [Huang et al. \(2021\)](#) does not predict which fam-
ily wins a scenario, consistent with its definition as a bound on the *potential* for a
separation rather than its sign; a large geometric difference is compatible with equal
or worse OOD accuracy, as we observe. That effective rank and OOD alignment are
associated with performance is a statement about kernels in general, not about quan-
tum kernels—which is precisely why the fidelity maps hold no privileged position: they
are one costly route into a geometric regime that a short-length-scale classical kernel
reaches directly.

The high-rank finding must also be reconciled with two spectral results that super-
ficially point the other way. Kübler et al. argue that quantum kernels generalize
in-distribution only when the relevant spectral mass is concentrated in few compo-
nents [Kübler et al. \(2021\)](#), and Thanasilp et al. show that fidelity kernels concentrate
exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit
counts grow [Thanasilp et al. \(2024\)](#). There is no contradiction, but there is a genuine
tension that our results locate precisely. Both results concern the asymptotic or i.i.d.
regime; our effective ranks (roughly 10–150 on training sets of 10^3 – 4×10^3 samples at
4–12 qubits) sit far below the concentration regime, so no contradiction arises. Our

finite-shot analysis, however, adds an empirical caveat these theoretical results anticipate: effective rank measured from finitely estimated fidelities inflates substantially (Section 2.6), so the exact effective ranks we report are a statevector idealization, and any account that leans on effective rank as the operative quantity would need many shots to reproduce even at the modest qubit counts studied here.

We emphasize what the geometry analysis does not deliver. It is a post-hoc association, not a deployable selection rule: choosing a configuration by training-set geometry alone does not reliably identify the better kernel, and OOD alignment requires OOD labels. Because the two families are statistically indistinguishable under honest selection, the more useful open problem is no longer “which family wins” but whether any label-free, train-only signal can select a robust kernel at all—a question that concerns classical and quantum kernels equally.

From a practical perspective, the paper suggests three takeaways for security ML. The first is substantive: on the evidence here, fidelity-based quantum kernels do not offer a robustness advantage under distribution shift over well-tuned classical kernels, while their pairwise overlap estimation introduces circuit and shot costs absent from direct evaluation of the classical baselines. A median-heuristic Laplacian kernel, especially one with a shortened length scale and its regularization tuned, is a strong and inexpensive robustness baseline, and it matches the fidelity maps under no-OOD-label selection on both modalities. Practitioners weighing a quantum-kernel pipeline for OOD-sensitive security detection should, on this evidence, prefer a tuned classical kernel unless a specific regime demonstrably favours the quantum construction after the same controls.

The second is methodological. Independently of the preferred kernel family, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed, and swap only the kernel;

2. define explicit OOD splits tied to interpretable shift mechanisms, including at least	737
one non-constructed shift mechanism where the data permit;	738
3. give every family the same freedom—the same candidate budget, symmetric length-	739
scale tuning, and per-configuration regularization tuned on training data alone;	740
4. select the deployed configuration without out-of-distribution labels, and keep an	741
in-distribution test split disjoint from the validation split used for selection;	742
5. repeat the pipeline, report variability as conditional intervals, and treat correlated	743
benchmarks as fixed case studies rather than pooling them into a single population	744
p -value.	745
	746
Security ML frequently suffers from comparisons in which model class, tuning effort,	747
and evaluation conditions change simultaneously; the way the verdict of Sections 2.2–	748
2.4 moves from a large apparent advantage to a null as the protocol is made fair is a	749
concrete demonstration of how much the conclusion depends on getting this right.	750
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The third concerns diagnostics: the geometry descriptors are cheap to compute	752
from Gram matrices that the pipeline already builds, and the alignment-survival anal-	753
ysis can be run post hoc on any deployed kernel system with a labeled drift sample.	754
Practitioners should nonetheless resist reading these results as a blanket recommen-	755
dation to replace classical kernels; the responsible recommendation is to test the	756
candidate families under the expected shift regime, using exactly the kind of protocol	757
described here.	758
	759
Several limitations define the scope of our conclusions.	760
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Our shift operationalization covers three reproducible mechanisms—within-class	762
sparsity-tail extremes, train-geometry-tail extremes, and a held-out attack-campaign	763
shift on network traffic (an unseen attack campaign plus a capture-partition change,	764
not a timestamped temporal split)—but not the full space of real-world drift processes.	765
Explicit temporal splits on timestamped malware, family-based drift, and adversarially	766
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783 induced shift remain outside the study’s scope, and our conclusions should be re-tested
784 under them rather than assumed.
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786 Kernel and encoding coverage also remains finite. The classical family now spans
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788 seven kernels plus a bandwidth sweep, and both families receive symmetric tuning
789 freedom, but neither candidate set is exhaustive: the quantum side is restricted to four
790 commonly used fidelity feature maps at low qubit counts, and trained or task-adapted
791 encodings, projected kernels, and deeper maps are not evaluated. The mechanism gives
792 a concrete prediction for such extensions—what should matter is their Gram-matrix
793 geometry, not their provenance—which future work can falsify. At substantially larger
794 qubit counts, exponential kernel concentration [Thanasilp et al. \(2024\)](#) would erode
795 the alignment-preserving rank the mechanism rewards, so our conclusions should not
796 be extrapolated beyond the low-qubit regime studied here.
797

803 Selection is necessarily over finite candidate sets. P1’ (ID-validation selection),
804 P2 (cross-realization OOD-supervised selection), and P3 (same-test OOD oracle) are
805 selection procedures over finite candidate sets. The primary comparison matches can-
806 didate budgets, and the full-pool and budget curves provide sensitivities, but residual
807 dependence on candidate definitions and search granularity remains. Fully nested
808 selection over additional independent benchmark pools would strengthen the analysis
809 further.
810

814 The quantum kernels are simulated rather than executed on hardware. The study
815 therefore characterizes the geometry of fidelity kernels, not the behavior of noisy
816 devices: finite sampling of kernel entries, hardware noise, and error mitigation would all
817 perturb the Gram matrices [Wang et al. \(2021\)](#), and exponential concentration implies
818 that with polynomially many shots the very spectral richness our mechanism rewards
819 becomes expensive to resolve at larger scales [Thanasilp et al. \(2024\)](#); the effect of either
820 on alignment survival is unknown and worth measuring. We make no claim about
821 hardware feasibility, resource advantage, or quantum speedup—the only rigorously
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established quantum-classical learning separation concerns a specifically constructed problem, not natural data [Liu et al. \(2021\)](#). At the dimensions studied, the fidelity kernels are also classically simulable [Schreiber et al. \(2023\)](#); [Jerbi et al. \(2024\)](#), which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats provide a final boundary. Four benchmark scenarios across two modalities is a substantial widening relative to single-dataset studies, but security is broader still: dynamic malware analysis and encrypted traffic remain untested. Within the network scenarios, the feature sets inherited from the public benchmarks retain port numbers and protocol indicators, which are legitimate but known shortcut-prone features in NIDS benchmarks [Engelen et al. \(2021\)](#); because every kernel receives identical features, this affects absolute accuracies, not the family comparisons. Finally, the mechanism’s effective-rank component is itself regime-dependent (strong on EMBER and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than smooth over.

In light of these limitations, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or hardware relevance—nor do we claim to have refuted quantum kernels in general. We claim three narrower things, supported by conditional effect estimates rather than a population p -value: that the apparent out-of-distribution advantage of fidelity kernels over customary baselines disappears when same-test oracle selection and untuned regularization are removed; that under equal candidate budgets and no-OOD-label selection no robust family-level advantage remains against either the customary or the extended classical reference, across eight fixed scenario-groups and two classifiers; and that the geometric descriptors associated with out-of-distribution behaviour occur in both families and do not identify kernel provenance as the source of robustness. This view is consistent with recent calls for careful benchmarking in QML [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#); [Kakavand](#)

875 [et al. \(2026\)](#) and is more informative than either an unqualified advantage claim or an
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877 unqualified dismissal: it identifies the evaluation choices that generate the optimistic
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879 result and shows how the conclusion changes when they are controlled.

880 This paper asked why kernel families behave differently under distribution shift,
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882 using the quantum-versus-classical comparison as both motivation and stress test. A
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884 controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and
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886 tuning freedom held fixed while only the kernel varies—was instantiated on four
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888 security scenarios spanning two modalities and three shift mechanisms, two clas-
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890 sifier families including a probabilistic one, and 72 principal settings with fifteen
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892 pipeline realizations. Effects are reported as fixed-case summaries with conditional
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894 pipeline-realization intervals, not as population significance tests.

895 The empirical arc has three movements. Under same-test oracle selection and fixed
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897 regularization, fidelity-based quantum kernels appear to beat the customary linear and
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899 RBF baselines under shift. That result does not survive the controlled comparison:
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901 once each configuration is tuned on training data, selection uses no OOD labels, and
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903 both extended families search an equal budget of kernel geometries, the dataset-equal-
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905 weighted OOD difference is -0.005 under the primary SVC analysis and -0.002 under
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907 the GP classifier. The primary comparison is classical-favoured or tied in seven of eight
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909 scenario-groups. At matched effective rank the classical kernels are at least as accurate
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911 everywhere. Effective rank and OOD alignment are associated with performance in a
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913 regime-dependent way, are shared by both families, and, a finite-shot analysis shows,
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915 do not transfer unchanged from exact statevectors to finite measurement.

916 The contribution is therefore threefold. Empirically, the study provides a con-
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918 trolled, variability-aware comparison of quantum and classical kernels under shift
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920 whose central finding is a clean negative obtained under fair conditions: no robust
family-level advantage survives budget matching, regularization tuning, and honest
selection. Methodologically, it isolates the specific evaluation choices—oracle selection,

fixed regularization, mismatched budgets, and pooled significance testing over correlated settings—that manufacture apparent quantum-kernel advantages, and it leaves behind a reproducible protocol that removes each. Analytically, it shows that the geometric quantities often invoked to explain such advantages describe both families equally and do not transfer unchanged to finite-shot estimation.

The most useful open problem this leaves is not “which family wins” but whether any label-free, train-only signal can select a robust kernel under shift at all; the security community’s label-free drift detectors [Yang et al. \(2021\)](#) and the theory of task–model alignment [Bordelon et al. \(2020\)](#); [Canatar et al. \(2021\)](#); [Ma et al. \(2023\)](#) are natural starting points. It is a question about kernels in general, and answering it would benefit classical and quantum methods alike—which is, in the end, the point: the productive question is not whether one family dominates, but which similarity geometry aligns with the shift regime one actually faces, and how to find it without peeking.

4 Methods

4.1 Problem setting

We study binary security classification—malware versus benign software, and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution data-generating process and D_{OOD} an out-of-distribution process induced by a constructed stress test or a held-out attack campaign. Given training data $T \sim D_{\text{ID}}$, we evaluate on an ID test set S_{ID} and an OOD test set S_{OOD} .

To isolate the kernel, classifier, preprocessing, optimization protocol, and split design are fixed. Classical and quantum Gram matrices are therefore consumed by the same downstream learner on the same samples.

967 4.2 Datasets and split construction

968
969 The four scenarios span static malware (EMBER) [Anderson and Roth \(2018\)](#) and
970 network flows: UNSW-NB15 DoS, UNSW-NB15 Reconnaissance [Moustafa and Slay](#)
971 [\(2015\)](#), and ToN-IoT Scanning [Alsaedi et al. \(2020\)](#). Every setting has shared T , S_{ID} ,
972 and S_{OOD} samples with identical class balance across kernel families.
973

974 EMBER uses two label-conditional tail stress tests. The $m1$ split holds out within-
975 class extremes of feature-block sparsity; $m2$ holds out within-class extremes of distance
976 to train-fitted class centroids. Network scenarios use an analogous train-centroid con-
977 struction and a held-out attack-campaign regime. In the latter, training and ID pools
978 contain benign traffic plus non-target attacks, whereas OOD contains benign traffic
979 plus the target campaign; UNSW pools additionally differ in capture partition. These
980 are reproducible stress tests, not timestamped temporal splits or certified covariate
981 shifts. Exact algorithms, feature exclusions, leakage checks, and pool definitions are
982 in the Supplementary Note “Dataset and shift construction.”
983

992 4.3 Controlled kernel-swap protocol

993
994 Figure 1 summarizes the controlled pipeline. Both SVC and a binary Gaussian-process
995 classifier with a logistic-link Laplace approximation consume unit-diagonal precom-
996 puted Gram matrices [Cortes and Vapnik \(1995\)](#); [Rasmussen and Williams \(2006\)](#).
997 Comparisons are always within a classifier. The GP classifier supplies approximate
998 probabilities for uncertainty diagnostics; no post-hoc calibration is applied.
999

1000 Training-fitted MaxAbs scaling and TruncatedSVD produce shared representations
1001 at $d \in \{4, 6, 8, 10, 12\}$. Quantum inputs are then mapped linearly to $[0, \pi]$. Dimension
1002 is an explicit candidate axis.
1003

4.4 Kernel families

The customary classical reference contains linear and RBF kernels. The extended reference adds degree-2 and degree-3 polynomial, Laplacian, and Matérn-3/2 and 5/2 kernels. RBF, Laplacian, and Matérn bandwidths sweep factors $\{0.1, 0.3, 1, 3, 10\}$ around training-only scale or median-distance heuristics. The quantum family uses fidelity

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2$$

for two ZZ depths, PauliXZ, and Z feature maps [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#). Every map sweeps angle scales $\{0.5, 1, 2\}$. Classical bandwidth and quantum angle-scale freedom are symmetric in purpose [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#). Exact formulas and grids appear in the Supplementary Note “Controlled kernel-swap protocol.”

4.5 Selection, candidate budgets, and inference

For each geometry, SVC regularization $C \in \{0.01, 0.1, 1, 10, 100\}$ is selected by five-fold stratified cross-validation on the training Gram matrix. The primary P1’ protocol divides the ID holdout into disjoint validation and test halves. Within each family, the candidate with the highest ID-validation balanced accuracy is deployed on ID-test and OOD-test; OOD labels never enter this choice. Same-test OOD-oracle and cross-realization OOD-supervised protocols are descriptive sensitivities.

Best-of-family selection favours a larger pool. The primary extended-family endpoint therefore uses 60 candidates per family where coverage is complete and 20 where it is reduced, generated by 5,000 classical-pool subsamples under three schemes. The full 115-candidate classical pool is secondary.

Every configuration has fifteen pipeline realizations (five split seeds by three model seeds). Two EMBER and six network scenario-groups are crossed with three master seeds and $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$ regimes (1000, 500, 500), (2000, 1000, 1000), and

(4000, 1800, 1800). Because scenario-groups share source pools and are not exchangeable, they are fixed case studies. Per group we average over the five split-seed clusters and report a conditional interval; across groups we use dataset-equal means and leave-one-dataset-out ranges. We attach no population p -value to eight fixed, correlated cases. The complete grid is in Supplementary Table 1 and resampling details in the Supplementary Note “Controlled kernel-swap protocol.”

4.6 Specification curve

The specification curve is a descriptive multiverse over every logically valid endpoint for which the frozen legacy and v4 artifacts contain complete paired data. Its ten specifications cross, where available, legacy or v4 generation; same-test OOD-oracle, legacy ID-based, or disjoint ID-validation selection; fixed $C = 1$ or five-fold train-CV regularization for SVC; customary linear/RBF or extended classical references; and native or equal candidate budgets. The order S1–S10 was fixed before aggregation and is never sorted by effect size. Within each specification, pipeline realizations are first averaged within setting and then within scenario-group; groups are averaged within source dataset before the three datasets receive equal weight. All eight group effects are displayed, and no specification-level hypothesis test is performed. Exact definitions and machine-readable group- and dataset-level values are provided in the Supplementary Information and versioned result artifact.

4.7 Prospectively frozen external validation

Before acquiring or inspecting any external quantum–classical result, we froze the tasks, sampling, preprocessing, candidate pools, estimands, uncertainty procedure, and interpretation rule in a version-controlled protocol. The external source is TableShift, a benchmark of real-world distribution shifts in tabular binary classification [Gardner et al. \(2023\)](#). We fixed three public tasks from distinct domains and with distinct

published split variables: College Scorecard (institution type), diabetes readmission (admission source), and ACS income (geographic region). TableShift’s predefined `train`, `validation`, `id_test`, and `ood_test` roles are preserved; its `ood_validation` split is excluded from preprocessing, tuning, selection, and reporting.

Each task has two nested sample-size strata: $(n_{\text{train}}, n_{\text{val}}, n_{\text{ID}}, n_{\text{OOD}}) = (1000, 250, 250, 500)$ and $(2000, 500, 500, 1000)$. Five label-blind deterministic subsampling seeds form the conditional resampling clusters. Training-only column filtering, numeric median and categorical mode imputation, one-hot encoding, MaxAbs scaling, TruncatedSVD, standardization, and range mapping produce the same dimensions and kernel pools as in the security study. The primary deployed configuration is selected on `validation`; `ood_test` is touched only for final evaluation. The same-test OOD oracle is retained as a non-deployable diagnostic.

For each task, size, seed, and classifier, the 60-candidate quantum pool is compared with 5,000 equal-size, whole-kernel-block subsamples of the 115-candidate classical pool. The external primary effect is the selected quantum OOD accuracy minus the expected selected classical OOD accuracy over these budget-matched subsamples. The protocol-contraction diagnostic subtracts this controlled effect from the weak-reference, fixed- C , same-test-oracle effect. We average seeds within task and size, give the two nested sizes equal weight within task, and give the three fixed tasks equal weight. Conditional 95% intervals use 9,999 resamples of seed clusters within task; no population p -value is attached to the three deliberately selected tasks. Full acquisition provenance, integrity gates, and the frozen outcome-classification rule are reported in the Supplementary Information.

4.8 Metrics and geometry analysis

OOD balanced accuracy is primary. Family effects are quantum minus classical after selection. ID balanced accuracy and ID-to-OOD degradation are secondary. For the

GP classifier we also record log loss, Brier score, 15-bin expected calibration error [Guo et al. \(2017\)](#), and predictive entropy.

From the classifier Gram matrices we compute effective rank [Roy and Vetterli \(2007\)](#), centered kernel-target alignment [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#), entry concentration, cross-split similarity, and regularized geometric difference [Huang et al. \(2021\)](#). We relate geometry to OOD accuracy within scenario-groups, use partial association to control family and length scale, test portability by leaving one dataset out, and pair quantum candidates with nearest-effective-rank classical candidates within run and dimension. These analyses are associational. Formulas and diagnostic definitions appear in the Supplementary Note “Metrics and geometry descriptors.”

4.9 Finite-shot perturbation

Exact fidelity entries are replaced by binomial estimates at $s \in \{128, 512, 2048, 8192\}$ shots. Training blocks are symmetrized, clipped, set to unit diagonal, and projected to the positive-semidefinite cone. The model isolates sampling error, not device noise. Projection diagnostics and interpretation are in the Supplementary Note “Finite-shot fidelity-estimation model.”

4.10 Use of generative AI

OpenAI Codex was used during manuscript revision to assist with language editing, code inspection, and consistency checking. The authors reviewed and take responsibility for all analyses, claims, citations, and text; the system is not an author.

Declarations

Data availability. This study uses the public EMBER [Anderson and Roth \(2018\)](#), UNSW-NB15 [Moustafa and Slay \(2015\)](#), ToN-IoT [Alsaedi et al. \(2020\)](#), and TableShift [Gardner et al. \(2023\)](#) benchmarks, which remain subject to their original access

conditions. Raw TableShift records are not redistributed. Derived split definitions, deterministic row-position hashes, shift constructions, experiment manifests, aggregated results, and summary tables are available in the Zenodo repository at <https://doi.org/10.5281/zenodo.19147649>. Before submission, this DOI will be verified against the exact archived release used to build the manuscript.

Code availability. Code for constructing the splits, running the kernel experiments, reproducing the confirmatory analysis, and regenerating every table and figure is available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework and in the versioned Zenodo archive above. The platform-independent Python 3.12 environment is pinned in `environment.lock.yml`; the code is released under BSD-3-Clause with no restriction on non-academic use.

Competing interests. The authors declare that they have no competing interests.

Authors' contributions. RFB: conceptualization, methodology, software, data curation, formal analysis, visualization, investigation, writing—original draft, writing—review and editing. IPL: supervision, validation, writing—review and editing. AGS: validation, project administration, writing—review and editing. PGB: supervision, resources, writing—review and editing. All authors read and approved the final manuscript.

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